

Ecosystems, Organisms, and Machines

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My theme is the concept, and the term, "self-organization." The history of this term, originally introduced by Immanuel Kant to characterize the unique properties of living organisms, is inseparable from the history of biology. Only in the second half of the 20th century does it begin to acquire the promise of a physicalistic understanding. This it does with two critical transformations in the meaning of the term: first, with the advent of cybernetics and its dissolution of the boundary between organisms and machines, and second, with the mathematical triumphs of nonlinear dynamical systems theory and accompanying claims to having dissolved the boundary between organisms and such physical phenomena as thunderstorms. How do these transformations affect the applicability of self-organization to the ecosystem—that provocatively hybrid entity that is part organism, part machine, and perhaps even part thunderstorm?

Keywords: self-organization, ecosystem, organism, machine

I am interested in words, and I approach my topic—the interaction of ecosystems, organisms, and machines—through an investigation of the term "self-organization." A good place to begin is by asking what it means. But, as is so often the case for important words in science, the answer is not simple. It turns out that the term "self-organization" is even more difficult to pin down than "ecosystem," and subject to such vagary as to lead some to suggest it is an empty phrase. The term has an interesting history of meanings and associations, however, which I want to review, looking at the changes in meaning in relation to organisms and to machines. Then I will return to the question of ecosystems, and their relation to organisms, machines, and the concept of self-organization.

Perhaps the most conspicuous feature of the history of the term "self-organization" is its inseparability from the history of biology—that is, from the history of the science concerned with the problem of life, which, as such, has been explicitly demarcated from other sciences only since 1800. The term was originally introduced in an effort to address the fundamental question of the life sciences, namely, What is an organism? or, put another way, What is the special property or feature that distinguishes a living system from a collection of inanimate matter? This was the question that first defined biology

as a separate and distinctive science at the beginning of the 19th century. And by its phrasing (implicit in the root meaning of the word *organism*), it specified at least the form of what would count as an answer. For what led to the common grouping of plants and animals in the first place—that is, what makes "the two genres of organized beings" (as Buffon referred to them) *organisms*—was a new focus on their conspicuous property of being organized, and of being organized in a particular way. As François Jacob observes, by the end of the 18th century, it was "by its organization [that] the living could be distinguished from the non-living.... Organization assembled the parts of the organism into a whole, enabled it to cope with the demands of life and imposed forms throughout the living world" (Jacob 1973, p. 74).

Only by that special arrangement and interaction of parts that brings the wellsprings of form and behavior of an organism *inside* itself could one distinguish an organism from its Greek root, *organon*, or tool. A tool, of necessity, requires a tool user, whereas an organism is a system of organs (or tools) that behaves as if it had a mind of its own, that is, it governs itself.

Indeed, the two words *organism* and *organization* acquired their contemporary usage more or less contemporaneously. Immanuel Kant, in 1790, gave one of the first modern definitions of an organism—not as a definition per se, but rather as a principle or "maxim" that, he wrote, "serves to define what is meant as an organism": namely, "an organized natural

Note: Given that this article appears in a special section on "ecology, complexity, and metaphor," some explanation may be required for the fact that I include no discussion of metaphor—of what it means or of how it works—and indeed, only once does the word even appear. There is a reason for this: for a term to be used metaphorically, it must first have a reference that is quite concrete or literal. In my view, "self-organization" lacks the requisite concreteness to qualify as a metaphor, and the same is surely true of "complexity." Insofar as my central concern is with the changing meanings of "self-organization," this essay might be read as an argument for why this term does not qualify.

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product is one in which every part is reciprocally both end and means. In such a product nothing is in vain, without an end, or to be ascribed to a blind mechanism of nature" (Kant 1993, p. 558). Organisms, he wrote, are the beings that "first afford objective reality to the conception of an *end* that is an end of *nature* and not a practical end. They supply natural science with the basis for a teleology...that would otherwise be absolutely unjustifiable to introduce into that science—seeing that we are quite unable to perceive a priori the possibility of such a kind of causality" (Kant 1993, p. 558).

Elaborating on this kind of causality, Kant wrote:

In such a natural product as this every part is thought as *owing* its presence to the agency of all the remaining parts, and also as existing *for the sake of the others* and of the whole, that is as an instrument, or organ.... The part must be an organ *producing* the other parts—each, consequently, reciprocally producing the others.... Only under these conditions and upon these terms can such a product be an *organized* and *self-organized being*, and, as such, be called a *physical end*. (Kant 1993, p. 557)

Indeed, it is here that the term "self-organized" first makes its appearance in relation to living beings. For Kant, the concept of self-organization liberates living creatures from the necessity for a designer. No external force, no divine architect, is responsible for the organization of nature, only the internal dynamics of the being itself.

The beginnings of biology thus prescribed not only the subject and primary question of the new science but also the form of answer to be sought. To say what an organism is would be to describe and delineate the particular character of the organization that defined its inner purposiveness, that gave it a mind of its own, that enabled it to organize itself. What is an organism? It is a bounded body capable not only of self-regulation and self-steering but also, and perhaps most important, of self-formation and self-generation. An organism is a body that, by virtue of its peculiar and particular organization, is made into a "self" which, even though not hermetically sealed (or perhaps because it is not hermetically sealed), achieves autonomy and the capacity for self-generation. The obvious task for biology was to understand the character of this special kind of organization or self-organization. At the close of the 18th century and the dawn of the 19th, it was evident—to Kant, as to his contemporaries—that neither blind chance nor mere mechanism, and certainly no machine that was then available, could suffice. "Strictly speaking," Kant wrote, "the organization of nature has nothing analogous to any causality known to us" (Kant 1993, p. 557). The science of such a mechanism would have to be a new kind of science, one that, not surprisingly (especially given the technology of his age), Kant assumed to be irreducible to physics and chemistry (see Keller 2004 for further discussion).

Admittedly, demarcating organisms in the real world was not always easy. For instance, the kinds of organization that

the "social organisms" gave rise to posed a particular challenge. Should we think of ant or termite colonies as organisms? Beehives? Coral communities? Humans too are social organisms—should the societies they form be regarded as organisms in and of themselves? Aren't they also self-organizing? And what about natural communities more generally? Is it useful to think of them, as Frederic Clements proposed in 1916, as constituting a "complex organism," governed by laws of development?

Indeed, metaphors of organism and also, albeit to a lesser extent, of self-organization ran rampant throughout the biological and social sciences. Wherever they were applied, the terms *organism* and *self-organization*, even in the absence of explicit criteria for delineation, remained tightly linked and, until the 1940s, set apart from the realm of inanimate objects, especially from those objects that were designed and built to serve human goals—that is, machines. Machines were designed, and they were designed from without. Of course, talk of design never did disappear from biology, but here the design in question—the organization—was recognized as internally generated. The burden of the concept of self-organization thus fell on the term *self*, for it is the self as source of organization that prevents an organism from ever being confused with a machine. Self-organization was the unmistakable and inimitable signature of living systems.

The first major mutation in this tradition came during World War II. "Out of the wickedness of war," as Warren Weaver put it, emerged not only a new machine but a new vision of a science of the inanimate: a science based on principles of feedback and circular causality, and aimed at the mechanical implementation of exactly the kind of purposive "organized complexity" so vividly exemplified by biological organisms (Weaver 1948)—in other words, a science that would repudiate the very distinction between organism and machine on which the concept of self-organization was predicated. Norbert Wiener called this new science *cybernetics*, and his vision quickly attracted the interest of a motley crew of mathematicians, physicists, and engineers (Wiener 1948). Ross Ashby was a particularly important influence in this new field. In the 1940s, he built a machine (the homeostat) that was to serve as a primitive model for a brain. Ashby's homeostat showed that a machine, properly designed, could exhibit autonomous, self-organizing behavior. And this, together with his work on "principles of the self-organizing dynamic system," served as a major stimulus for the rash of conferences on self-organizing systems organized by the Office of Naval Research (ONR) in the 1950s (Ashby 1947). All of these discussions and investigations were conducted with an explicit engineering goal, namely, the design and construction of systems that could organize themselves, grow themselves, and perhaps even reproduce themselves. The crucial property of these systems (see, e.g., Ashby 1962, Pask 1992) lay in the relations between components; it was out of these relations that functionality or purpose would emerge. The goal was to understand the logic of the coupling (or "conversation") between subsystems that would generate the capacity of such

a system to have its own objectives, that is, to be a natural system.

Frank Rosenblatt's Perceptron project embodied this vision in its most ambitious form, but despite enthusiastic support in its early stages from the ONR, in the end the Perceptron failed to produce anything approximating a live tissue. By the 1960s, support for the project had dried up, and the spirit of artificial intelligence, organized around very different principles, had moved elsewhere. For all their enthusiasm and inventiveness, the program that Ashby, Rosenblatt, and their colleagues sought to build, and their efforts to establish self-organization as a general concept capable of unifying the animate and inanimate worlds, did not succeed (at least in the short run) in capturing the imagination either of the public or of the larger scientific community. But they can claim one success, and it is a significant one. After their efforts, self-organization could no longer be seen as an exclusively biological concept; confidence in the solidity of the boundary between organisms and machines was now irrevocably eroded.

The next important mutation in this history came with the emergence of studies of nonlinear dynamical systems in physics and mathematics in the late 1970s and 1980s, and it was with this mutation that the term "self-organization" captured the imagination of both the public and the wider scientific community. In this resurfacing, the term underwent a further shift in meaning. For Ilya Prigogine (1969), self-organization referred to the emergence of "dissipative structures" in systems far from equilibrium and low in entropy. Given a sufficiently large flux of free energy in such systems, one can expect to see the spontaneous emergence of such striking phenomena as eddies, vortices, or Bénard rolls and cells. At roughly the same time, the study of nonlinear dynamical systems by mathematicians was yielding similar insights about these same phenomena, described in terms of stable attractors and limit cycles. For both mathematicians and physicists, self-organization now referred to the production of stable patterns observed in physical, and sometimes in biological, systems governed by nonlinear dynamics. Enthusiasm for such analyses ran high, generating some extraordinary expectations. As Paul Davies wrote, "Mathematically we can now see how nonlinearity in far-from-equilibrium systems can induce matter [here quoting Bennett 1986] to 'transcend the clod-like nature it would manifest at equilibrium, and behave instead in dramatic and unforeseen ways, molding itself for example into thunderstorms, people and umbrellas'" (Davies 1989, p. 111).

The late 1980s brought a noteworthy addition to these ideas, namely, Per Bak's notion of "self-organized criticality" (SOC; Bak et al. 1987). In 1996, Bak reported that, since the coining of the phrase in 1987, "more than 2,000 papers have been written on the subject [of SOC]," making Bak's initial

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paper "the most cited in physics" (Bak 1996). Stuart Kauffman's 1993 book, *The Origins of Order* (which Bak referred to as "the first serious attempt to model a complete biology"), sought to apply Bak's ideas to evolution; but Kauffman's vision of evolution operating on "interacting dancing landscapes," which, driven by self-organizing dynamics, would approach criticality, turned out to be premature. Almost immediately afterward, however, Bak and Sneppen (1993) succeeded in developing a model for evolutionary dynamics that exhibited all the desired characteristics. Theirs was a model of an open and dissipative system that organizes itself into a critical state by its intrinsic dynamics, independent of any control parameter. Drawing on an analogy with the physics of phase transitions, the existence of a critical state is said to be signaled by a power-law distribution in some variable—a clear indication, to physicists familiar with the behavior of systems at thermodynamic critical points, that short-range interactions have induced a kind of global organization in which details of the particular system are obliterated. The familiar image is that of sand being dripped onto a sand pile. Once the sand pile has attained a critical slope, it retains its conical shape as more sand is added; it manages this by setting off small avalanches. The timing and size of individual avalanches are unpredictable, but the distribution of avalanches (in both size and timing) displays the prototypical regularity of the power law.

Self-organized criticality soon became a new buzzword, the latest key to an understanding of emerging structure in complex systems. According to Bak and his colleagues,

The dynamics of complex systems in nature often occurs in terms of punctuations, or avalanches, rather than following a smooth gradual path. Earthquakes, stock market crashes, and mass extinctions in biology are examples of this turbulent behavior. It will be argued that this is a consequence of self-organized criticality, the process by which large dynamical systems evolve to a state with no characteristic time or length scale. (Paczuski et al. 1996)

Just a few years later, Laszlo Barabasi and his colleagues would extend Bak's idea of SOC to the world of network topology, providing yet another boost to expectations about what the physics of phase transitions might do for biology.

Here is a literature that began in the world of physics, written by physicists and published in physics journals, but rapidly spread to other fields, and was soon taken up by a flourishing industry of science popularization. The term "self-organization," as used throughout this literature, is severed both from its original biological meaning and from its later engineering sense; stripped of all resonances of design

in either of the senses invoked in these disciplines; and appropriated, instead, for the categorization of complex phenomena arising out of random ensembles, essentially uniform distributions of simple physical entities. Not only eddies, whirlpools, and Bénard cells but also more dramatic eruptions, such as thunderstorms, earthquakes, and living organisms, are to be understood as arising from homogeneous gases, fluids, and lattices. Indeed, it is claimed that the emergence of life itself can be seen as a self-organized critical phenomenon. Life is incorporated not into the category of structurally complex "self-organizing" machines, as it had been in the 1950s and 1960s, but into the nonlinear dynamics of structurally simple, physical-chemical systems.

What has happened here? In this assimilation of life and familiar physical processes, is biology being reduced to physics, or is physics being revived by the infusion of life? Lee Smolin, one of the more thoughtful writers on this subject, argues that viewing the universe as a nonequilibrium, self-organizing system has many advantages: in particular, it allows for a world in which "a variety of improbable structures—and indeed life itself—exist permanently, without need of pilot or other external agent, [and] offers the possibility of constructing a scientific cosmology that is finally liberated from the crippling duality that lies behind Plato's myth" (Smolin 1997, p. 160).

But despite Smolin's caution, and for all his hopes, there is a serious elision here. Analyses of nonlinear dynamical systems clearly demonstrate the ease with which complexity can be generated, but such arguments fail to recognize a crucial distinction—one that Warren Weaver described in 1948 as a distinction between disorganized and organized complexity, but that we might characterize as a difference between the generation (or emergence) of complexity and its organization. In the words of John Mattick, "The problem is not how to generate complexity—that is easy—but rather how to control it to specify ordered [and, I would add, robust] trajectories that lead to highly organized and complex organisms" (Mattick 2004, pp. 317–318).

This is a crucial point for biology, and I will return to it later. But now I want to take a slight detour to ask how, and where, ecology fits into this history. I've already cited Clements's comparison of natural communities to organisms. Probably the next important development in the history of ecology, at least in the United States, is A. G. Tansley's (1935) concept of the ecosystem. Ecosystems are systems that incorporate living and nonliving elements, explicitly overriding the boundary between animate and inanimate. As Tansley wrote, "In an ecosystem the organism and the inorganic factors alike are components which are in a relatively stable dynamic equilibrium" (Tansley 1935, p. 306). He hypothesized the existence of "universal processes tending toward the creation of such equilibrated systems," citing succession and development as instances of such processes. This concept comes out of engineering, and it was not obvious to early users of the term

whether it made sense to think of ecosystems as organisms. Frank Golley refers to the ecosystem concept as "a machine theory applied to nature," but ecosystems continue to hover among and between all three of our figures, simultaneously bearing features of organisms, machines, and thunderstorms (Golley 1993). As such, they might be said to lend themselves easily to the boundary-blurring wrought by the new conceptions of self-organization.

The actual term "self-organization" doesn't appear in Evelyn Hutchinson's classic paper, "Circular Causal Systems in Ecology" (1948), but Hutchinson's arguments clearly drew on these recent developments in cybernetics. In particular, Hutchinson was interested in "natural circular systems" and the various self-correcting and self-regulating mechanisms that act to damp oscillations or otherwise maintain stability in the circular causal pathways that characterize complex ecosystems. In fact, self-organization did not enter the literature of ecology until the mid-1980s, when Howard Odum employed it in its cybernetic sense as a convenient shorthand for the kind of ecosystems analysis he had been doing for decades. But the

main engagement of ecology with notions of self-organization came in the 1990s, with Bak and his arguments for the importance of SOC. Indeed, Bak viewed ecosystems as ideal candidates for SOC, and he found confirmation in the presence of power-law distributions in the size of extinctions. Here was a way of making sense of ecosystem develop-

ment without invoking the guiding hand of God, of man, or even, for that matter, of natural selection. The influence of Bak's arguments can be judged by the pervasiveness of references to SOC in the literature of theoretical ecology today.

As Simon Levin and others point out, however, power-law distributions are everywhere. Furthermore, they are not in themselves indicative of any particular mechanism of organization, and the relevance of SOC to ecosystems remains to be determined. Levin remains open-minded on the question of the relevance of SOC to ecosystems, but clearly recognizes that the failure of SOC to consider the effects of evolution on ecosystem development is a major liability. I am rather more skeptical, in part because of the question of how complexity is to be organized, and in part because there are so many other concerns that are also excluded.

In addition to the inclusion of evolutionary forces, Levin—perhaps unwittingly—offers another way of improving on physicists' notion of self-organization when he considers the contents of his desk. He writes, "The contents of my desk may not seem to have much to do with ecosystems, but take another look. My office is indeed a self-organized system, with me at the center. It has more the element of design than do ecological systems, yet it still reflects a huge dose of chance and historical influence" (Levin 1999, p. 158).

Now, this might be something of a pun, for Levin is the self who is organizing (or not, as the case may be) his desk. However, it is hardly customary to include that supremely inten-

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tional agent, the human self, in our conception of a self-organized system, so perhaps he should have written, "My office is indeed a myself-organized system." Similarly, we tend overwhelmingly to exclude humans from our conception of ecosystem. But let's stop and ask, Why is that? Why is it that, in every conception of self-organization, the self that is the source of organization is, in contrast to the human self (or "myself"), without intentionality or agency? Furthermore, such exclusion seems necessary, for otherwise what would prevent us from describing the system composed of an engineer and his machine as a self-organized system? Or from describing an urban landscape as an ecosystem? Levin tries to avoid this problem by downplaying his own agency, by emphasizing the randomness of the processes underlying the design of his office (he is not really in charge). But this is just to sidestep the issue of whether agency, or intentional design, *can* be brought into our concept of a self-organized system, in whatever form that concept takes. To bring the engineer into the system, to put Levin in his office with his agency and intentionality intact, would be to confound the entire tradition that takes human agency or intentionality as *a priori* unnatural, and accordingly pits natural against artificial design.

But of course we are natural beings! As contemporary scientists clearly understand, humans are products of evolution, just like all other organisms. Nevertheless, tradition, even post-Darwinian tradition, excludes our doings from natural history. It may acknowledge our effects on the natural world—at times (though less often in recent times) even celebrate them—but these effects are treated as impingements, and never incorporated into our conception of self-organization. Yet to put this exclusion so baldly is to make its absurdity self-evident, and to invite us to challenge the entire tradition on which it rests.

So here is a proposal: Let's drop the question of intentionality (after all, this too, like free will, may turn out to be an illusion) and focus instead on agency—an attribute we clearly share with many if not with all other organisms, and one that is, both scientifically and philosophically, surely problem enough. And let's think of the machines we create simultaneously as extensions of ourselves (insofar as they enhance our functionality) and, at the same time, as effects of human agency on our environment. In other words, let's try thinking of ourselves in the same terms as we think of other organisms—organisms that shape their environment by their activities, and that build entities extending their functionality, which can accordingly be thought of as extensions of themselves. Consider, for example, beaver dams; bird nests; any of Scott Turner's wonderful examples of the tunnels built by earthworms that serve as accessory kidneys; the bubble gills built by aquatic beetles; the horn-shaped burrows of crickets that amplify song; or homeostatic termite mounds (Turner 2000). And now think of all these systems as self-organizing systems. That is to say, think of a system in which the entire system is shaped by the combined activities of all the individual components—activities that are generated inside individual

components, with effects manifested externally to themselves, but all the while remaining inside the composite self that defines the larger system. The modifications engendered by these activities need not be random (and generally will not be), but nor need they be aimed at the coherence or survival of the system. It's just that those modifications that do enhance the survival of the system will, by definition, persist. Indeed, such an argument would merely be an extension of the reasoning that has led Kauffman (1986, 1993) and Ulanowicz (2001) to attribute a kind of agency to closed autocatalytic systems undergoing selection for increasing effectiveness.

Such arguments may lead us to think of everything as a self-organizing system, but doing so need not be bad—as long as it comes with the understanding that the most interesting kinds of self-organizing systems are those that require the participation and interaction of many different kinds of selves. And herein—in the heterogeneity of selves—lies the rub; especially in the different kind of self, and different order of agency, that human participation ushers into ecosystems. For we humans have perfected agency to such a degree that the effects of our participation threaten a degree of destabilization far surpassing the power of all other species to destabilize their, and our, ecosystems. There is indeed cause for worry about our impact on our environment, but not because human intentionality is in a different realm from the agency of other organisms. Rather, the danger is that the highly perfected form of our agency makes it so much more consequential.

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